EXPERIMENT 1

IMPLEMENTATION OF CAESAR CIPHER

AIM : To implement the Caesar Cipher substitution technique using Python.

ALGORITHM :

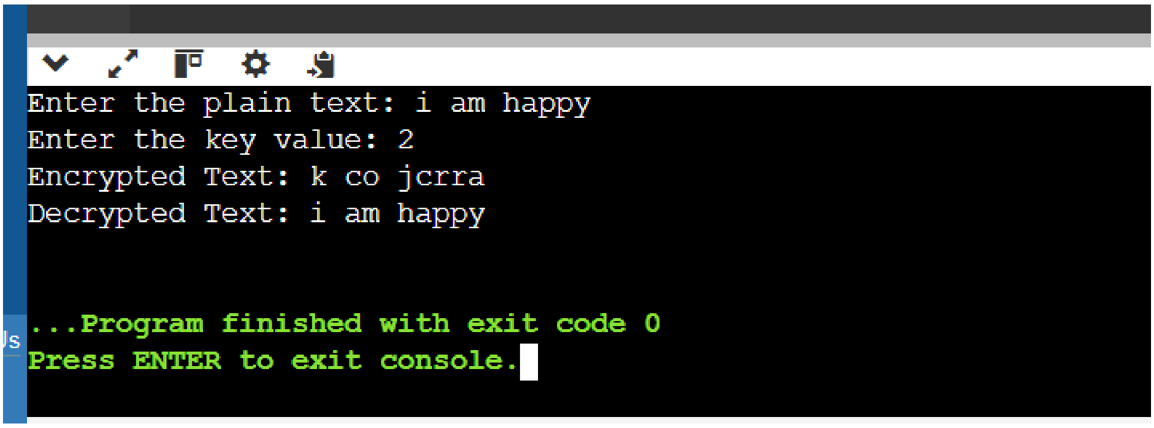
1. Read the plaintext from the user.
2. Read the key (shift value) from the user.
3. For each character in the plaintext:
    a. If it is an uppercase letter, shift it by the key within A-Z.
    b. If it is a lowercase letter, shift it by the key within a-z.
    c. Leave non-alphabetic characters unchanged.
4. Display the encrypted (cipher) text.
5. Decrypt the cipher text by shifting each character backward using the same key.
6. Display the decrypted text.

CODE:

```
def encrypt(text, key):
    cipher = ""
    for ch in text:
        if ch.isupper():
            cipher += chr((ord(ch) - ord('A') + key) % 26 + ord('A'))
        elif ch.islower():
            cipher += chr((ord(ch) - ord('a') + key) % 26 + ord('a'))
        else:
            cipher += ch
    return cipher
def decrypt(cipher, key):
    text = ""
```

```
    for ch in cipher:
        if ch.isupper():
            text += chr((ord(ch) - ord('A') - key) % 26 + ord('A'))
        elif ch.islower():
            text += chr((ord(ch) - ord('a') - key) % 26 + ord('a'))
        else:
            text += ch
    return text
plain = input("Enter the plain text: ")
key = int(input("Enter the key value: "))
cipher = encrypt(plain, key)
print("Encrypted Text:", cipher)
decrypted = decrypt(cipher, key)
print("Decrypted Text:", decrypted)
```

OUTPUT



RESULT

Thus, the Monoalphabetic Substitution Cipher was implemented successfully using Python.